

## Contact

[gabriel.unifei2017@gmail.com](mailto:gabriel.unifei2017@gmail.com)

[www.linkedin.com/in/gabriel-mouallem](https://www.linkedin.com/in/gabriel-mouallem) (LinkedIn)

## Top Skills

Banco de dados

Kubernetes

Node.js

# Gabriel Mouallem

Software & Product Engineer | Full Stack | Kubernetes (K8s) | React | Next.js | Python | TypeScript | Cloud Native | IoT | Testing Advocate  
Itajubá, Minas Gerais, Brazil

## Summary

Product/customer-oriented problem solver with ~6 years of experience specializing in full-stack development, with deep expertise in React/Next.js/TypeScript and Python (Django/FastAPI/Flask). Experienced across diverse domains including IoT systems, AI/Video analysis, and Cloud Infrastructure/DBaaS. Proven problem-solver, having previously led significant backend performance optimizations and developed complex offline-first mobile applications. Currently leading the technical development and product strategy for a Kubernetes-based PostgreSQL DBaaS (using CloudNativePG) at Latitude.sh. Passionate about delivering high-quality, user-centric solutions through robust engineering and testing.

---

## Experience

### Stealth

Product & Software Engineer

April 2026 - Present (2 months)

San Francisco Bay Area

Coming soon...

### Latitude.sh

1 year 11 months

Product & Software Engineer (Kubernetes)

February 2025 - April 2026 (1 year 3 months)

<https://latitude.sh>

- Transitioned to the Managed Services team (databases, virtual machines, filesystems) as the primary developer and product driver for Latitude.sh's Managed PostgreSQL DBaaS on bare-metal Kubernetes.
- Led technical design and delivery of DBaaS product features: automated S3 backups, connection pooling, IP whitelisting, audit logging, and customer-facing management tools.

- Implemented SOC 2 controls for data-at-rest and added comprehensive observability: Prometheus (metrics), Grafana (dashboards), Loki (logs), Alloy (tracing/alerts).
- Owned infrastructure versioning and multi-region replication strategy to support high availability and disaster recovery.
- Worked closely with CloudNativePG operator and Rancher API to run PostgreSQL at scale on Kubernetes, optimizing for performance and operational resilience.
- Explored CEPH for distributed storage as a foundation for future managed filesystem offerings.
- Continued to review and develop frontend DB features (admin, monitoring, UX flows), ensuring consistency between frontend product needs and backend infra capabilities.
- Drove product iteration: from MVP launch (300+ waitlist signups) to internal dogfooding and customer-informed refinement.

Key technologies & integrations: Kubernetes, Rancher API, CloudNativePG, PostgreSQL, Node.js, Git/GitHub, S3 backups, CEPH (intro), Prometheus, Grafana, Grafana Loki, Alloy.

#### Software Engineer (Frontend)

June 2024 - January 2025 (8 months)

<https://latitude.sh>

- Member of a high-bar product engineering team with strict standards for UX, UI, and copywriting, directly shaping the Latitude.sh customer-facing experience across marketing, product, and docs.
- Contributed across three frontend apps:
  - Website (landing pages) — implemented ContentLayer integrations and optimized UX/UI with a content-first approach.
- Dashboard (core customer platform) — primary area of ownership; delivered customer-facing features and platform flows; rapid MVPs validated via Next.js API routes, later hardened into a Node.js client library (for internal and external API consumption).
- Documentation site — integrated Mintlify for developer-focused, scalable docs.
- Ensured stability, monitoring, and observability of frontend services by:
  - Leveraging Vercel for deployments, environment management, edge routing, and firewall rules.
  - Bugsnag for proactive error/crash tracking, reducing production incidents.

- Segment.io as the analytics hub, routing telemetry into Google Tag Manager, Google Analytics, Posthog, Customer.io, and downstream tools.
- Championed a testing-first culture (Jest, React Testing Library, MSW) with Storybook-driven development and Turbopack, improving reliability and speeding up iterative delivery.
- Performed PR reviews on complex logic and design patterns, raising engineering standards for maintainability, accessibility, and UX consistency.
- Implemented security-conscious features (e.g., Iron Session encryption, WorkOS auth).

Key technologies & integrations: Next.js, React, TypeScript, Tailwind, Radix UI, shadcn/ui, SWR, Jest, RTL, MSW, Storybook, Turbopack, Vercel, Bugsnag, Segment.io, Posthog, Customer.io, Google Tag Manager, Google Analytics, Retool, Inngest, Algolia, WorkOS, Linear, Stripe, Sphere Pay, Contentful, ContentLayer, Mintlify.

**AFL Consultores Asociados**  
**Freelancer Fullstack Developer**  
 May 2024 - June 2024 (2 months)

- Accelerated project delivery through targeted full-stack support (React/TypeScript frontend, Python/FastAPI backend).
- Engineered reusable frontend components using advanced React patterns (Portals, Composition, Context API, custom hooks) to isolate business logic.
- Developed Python/FastAPI endpoints and implemented a SQLAlchemy data access layer to integrate with existing NestJS/Prisma services.
- Improved backend stability by writing Jest unit tests for NestJS modules/services.

Key Technologies: React, Typescript, React Query, Radix, shadcn, Tailwind CSS, Jest, React Testing Library, Python, FastAPI, SQLAlchemy, NestJS, Prisma.

**Turing**  
**Software Developer**  
 August 2022 - June 2024 (1 year 11 months)  
 Honolulu, Havai, Estados Unidos  
<https://ai.fish>

- Collaborated remotely (USD-based contract) with US client AI.FISH as the sole frontend developer, architecting and delivering 3 complex platforms for AI-powered video analysis.
- Engineered highly interactive frontend applications enabling users to analyze video streams integrated with real-time AI object detection results (e.g., accurately synchronized bounding boxes on moving video).
- Solved significant performance challenges by implementing sophisticated data fetching and caching strategies (interacting with the AI backend API) to ensure smooth playback and responsive scrubbing/seeking through long videos with extensive AI metadata.
- Designed a maintainable and scalable React frontend architecture using TypeScript, organizing features into distinct modules and co-locating component-specific files (types, constants, tests) to improve encapsulation and developer experience.
- Achieved 80% unit test coverage in a complex mocking environment using Jest, React Testing Library (RTL), and Mock Service Worker (MSW), ensuring high application reliability and enabling confident refactoring.

Key Technologies: Git, Github, Figma, React, Vite, Typescript, Jest, React Testing Library (RTL), MSW, React Query, Storybook, framer-motion, Legend App State, Material UI.

## VAAS - Transaction Monitoring Platform

Freelancer Frontend Developer

November 2023 - December 2023 (2 months)

Florianópolis, Santa Catarina, Brasil

<https://vaas.com.br>

- Engineered complex frontend dashboards for VAAS's cryptocurrency risk management platform (supporting AML, compliance, and security monitoring).
- Created dynamic and intricate data visualizations (treemaps, bar charts, etc.) using React, Next.js, TypeScript, and Apex Charts to provide actionable insights from transaction data.
- Executed advanced data manipulation and normalization logic purely on the frontend (TypeScript) to transform and prepare raw API data for complex charting requirements.

Key Technologies: React, Next.js, TypeScript, Apex Charts, Material-UI, Vercel, Context API, Axios, React-Query, @emotion/styled, @emotion/react.

## BEES

### Software Developer

September 2022 - August 2023 (1 year)

Campinas, São Paulo, Brasil

<https://mybeesapp.com>

- Collaborated remotely within a large-scale enterprise environment (BEES/AB-Inbev) on an internal platform managing multi-national product promotions and integrating with diverse country-specific ERP systems.
- Spearheaded a frontend testing initiative using BDD principles (Jest, React Testing Library) focused on UI behavior. Increased unit test coverage by 40% in 3 months, establishing a quality standard that later supported a company-wide 90% coverage requirement for deployments and significantly enhanced refactoring confidence.
- Designed and implemented the frontend interface for a critical new permission system. Collaborated closely with the backend team to model the system and ensure functional alignment, enabling granular access control based on country and user roles.
- Presented technical lectures on Redux to improve team understanding and adoption of state management best practices.

Key Technologies: Git, Azure, Figma, React, Typescript, Jest, React Testing Library, React Query, Redux, Redux Toolkit, Redux Saga.

## Accenture

### Senior Application Development Analyst

January 2022 - September 2022 (9 months)

São Paulo, Brasil

- Co-led the frontend migration of a large-scale CRM application (used across >30 countries) from legacy Ionic 3/Angular to React Native, successfully modernizing the mobile technology stack for Unilever.
- Engineered robust offline-first capabilities and managed complex data synchronization (offline-to-online) within the CRM using the Salesforce React Native SDK, ensuring reliable performance in varied connectivity scenarios.
- Served as a key senior technical lead (1 of 2), driving critical decisions regarding React Native architecture, state management patterns (Redux/Context), performance optimization strategies, and establishing coding standards.
- Mentored development team members through pair programming and technical guidance (Redux/Context); delivered impactful technical

presentations on React Native and performance to both the project team and large corporate audiences (hundreds of attendees) within Accenture Brazil.

- Saved significant development time by forking and tailoring a complex calendar UI component for specific drag-and-drop scheduling requirements, demonstrating proactive problem-solving.

Key Technologies: React Native, Typescript, Salesforce React Native SDK, Redux, Redux Saga, Redux Toolkit, Context API, Storybook, Azure, Git, Bitbucket, Ionic 3, Angular.

## ContraDP Monitores

Freelancer Frontend Developer

January 2022 - May 2022 (5 months)

Santa Catarina, Brasil

<https://contradp.com.br>

- Led the initial frontend development phase for the ContraDP monitoring platform and public landing page.
- Built foundational UI components and established the project structure using React, TypeScript, Chakra UI, and Emotion, adhering to modern development best practices.
- Implemented state management with Jotai and incorporated animations using Framer Motion to enhance user experience.

Key Technologies: React, Typescript, Chakra UI, Emotion, Framer Motion, Jotai, Heroku, Git.

## NTT DATA Europe & LATAM

Software Developer

November 2021 - January 2022 (3 months)

Uberlândia, Minas Gerais, Brasil

- Assigned to a Serasa Experian project developing a reusable report generation system using React, TypeScript, Node.js (Express/NestJS), and JSReport.

Key Technologies: Git, Bitbucket, Jira, React, Typescript, JSReport, Node.js (Express, NestJS), Docker.

## Irricontrol

2 years 5 months

## Full Stack Developer

September 2021 - November 2021 (3 months)

Itajubá, Minas Gerais, Brasil

- Provided informal technical leadership for a team of 3 developers, guiding implementations, onboarding new members, presenting system architecture, and offering technical support.
- Diagnosed and resolved critical backend performance bottlenecks that caused extremely slow initial page loads, achieving an approximate 1000% improvement in loading time.
- Implemented a multi-stage optimization strategy involving API pagination, Django Cache implementation, asynchronous background tasks (Celery), and foundational database schema refactoring (migrating models to abstract inheritance) to eliminate costly queries.
- Engaged directly with clients for technical support and participated actively in product development cycles, translating customer feedback into actionable technical requirements.
- Developed complex data visualization components using the JavaScript Canvas API for domain-specific UI features.

Key Technologies: Python, Django, Django REST Framework, Django Cache, Celery, PostgreSQL, WebSockets, React, Git, Heroku.

## Full Stack Developer Junior

January 2021 - September 2021 (9 months)

Itajubá, Minas Gerais, Brasil

- Managed application deployment, monitoring, and scaling on the Heroku platform and contributed to ongoing backend performance optimization initiatives.
- Implemented MQTT publish/subscribe logic enabling real-time communication and control for the company's core IoT solutions focused on Central Pivot irrigation systems.
- Co-developed low-level radio communication protocols, defining encrypted message structures mapped to device memory, supporting Over-the-Air (OTA) updates, and actively troubleshooting complex hardware/firmware/software version compatibility challenges.
- Acted as the primary technical liaison between hardware (embedded systems/C#) and software (Python/React) teams, analyzing requirements, assessing implementation feasibility, and identifying edge cases for new IoT features.

- Contributed key backend and frontend features for new product lines, including Level Meter monitoring and Irrigation Pump control systems, alongside enhancements for the flagship Central Pivot product.
- Assisted in onboarding and mentoring new developers through architectural presentations, infrastructure setup guidance, and pair/shadow programming.

Key Technologies: Python, Django, Django REST Framework, React, PostgreSQL, Heroku, MQTT, Git.

#### Full Stack Developer Intern

February 2020 - January 2021 (1 year)

Itajubá, Minas Gerais, Brasil

- Actively contributed to backend API development using Python and Django REST Framework, including implementing new endpoints, performing bug fixes, and assisting with general system maintenance.
- Generated complex, on-demand operational reports by performing intricate data manipulation and analysis with Pandas and NumPy (covering usage patterns, energy consumption, error rates, costs) to populate sophisticated Excel templates for internal BI and client insights.
- Implemented RecoilJS to refactor React application state management, significantly improving UI performance and reducing component complexity/prop drilling, especially beneficial for handling high-frequency WebSocket data updates.
- Gained foundational understanding of IoT systems and product lifecycles by participating actively in technical meetings discussing radio communication, MQTT, and firmware/hardware challenges.

Key Technologies: Python, Django, Django REST Framework, Pandas, NumPy, React, RecoilJS, WebSockets, Git, GitHub.

#### Mobile Developer Intern

July 2019 - February 2020 (8 months)

Itajubá, Minas Gerais

<https://irricontrol.com.br>

- Contributed to mobile app development (Ionic 3/Cordova with native Android Java/Kotlin) at an innovative IoT startup during its accelerator/incubator phase, performing maintenance and bug fixes.
- Developed and implemented a critical mobile alarm system for irrigation equipment failures, directly contributing to significant savings (estimated potentially in millions BRL) for clients by preventing water and resource waste.



- Played a key role in the complex migration of the mobile platform from Ionic 3 (Angular/Cordova) to Ionic 4 (React/Capacitor), successfully establishing React as the foundational technology for future frontend development within the company.

Key Technologies: Ionic 3, Ionic 4, React, Angular, Cordova, Capacitor, TypeScript, JavaScript, Java, Kotlin (Android), Redux, Context API, Axios, Socket.io, Google Maps API, Git.

---

## Education

Universidade Anhembi Morumbi

Bacharelado, Ciência da Computação · (February 2023 - February 2025)

Universidade Federal de Itajubá

Computer Science, Sem especialização · (January 2017 - July 2022)